OFS PROTOCOL TEMPLATE

Review title

Circulating Biomarkers of Cellular Senescence and Chemotherapy-Related Cognitive Impairment in Breast Cancer: A Scoping Review

General Data:

Authors and affiliations:
- Rodrigo Maldonado, PhD , Universidad Finis Terrae.
- Sebastian Pizarro, Mg Salud Pública, Universidad Finis Terrae.
- Constanza Lagos, estudiante de medicina en la universidad Finis Terrae.
- Javier Diaz, estudiante de medicina, clínica nurse, Mg Salud Pública, Universidad Finis Terrae.
- Ma Jesus, clinical nurse, Mg Investigacion en Ciencias de la Salud, Pontificia Universidad Católica de Chile.
- Corresponding Author:
  - Rodrigo Maldonado, PhD , Universidad Finis Terrae.
  - Contact mail: rmaldonado@uft.cl
- Conflicts of Interest:
- Funding / Support:
  - No funding needed.

Introduction and Justification

Background:
- Systemic cytotoxic chemotherapy used to treat breast cancer can induce therapy-induced cellular senescence in malignant and non-malignant tissues. Senescent cells undergo stable cell-cycle arrest and acquire a senescence-associated secretory phenotype (SASP), characterised by sustained secretion of pro-inflammatory cytokines, chemokines and tissue-remodelling factors.
- Circulating markers of senescence and SASP, measured in blood or peripheral blood cells, have been proposed as indicators of accelerated biological ageing and as potential contributors to neurotoxic pathways. These include p16INK4a (CDKN2A), IL-6, IL-8, IL-1β, TNF-α and GDF-15.
- Chemotherapy-related cognitive impairment (CRCI, "chemobrain") affects executive function, processing speed, attention and memory, and is assessed with objective neuropsychological batteries or validated patient-reported measures such as FACT-Cog.

Rationale:
- The evidence linking senescence biomarkers to CRCI is expected to be heterogeneous in biomarkers, matrices, assays, cognitive instruments and time points. A scoping review is therefore appropriate to map the extent and types of evidence, clarify concepts and identify gaps, rather than to estimate effects or pool results.

Review Question:
- What is known about the relationship between circulating biomarkers of

cellular senescence and SASP and chemotherapy-related cognitive impairment in adults with non-metastatic breast cancer exposed to cytotoxic chemotherapy?

Objectives: To map the evidence on circulating biomarkers of cellular senescence and SASP in relation to chemotherapy-related cognitive impairment in breast cancer.

- Identify and characterise the biomarkers, matrices and assays used.
- Describe cognitive outcomes and time points
- Summarise reported associations by regimen and time window
- Identify gasps for future research

Methodology

The review follows the JBI methodology for scoping reviews and will be reported according to PRISMA-ScR

Eligibility Criteria: Detailed inclusion and exclusion criteria based on:

- Participants / Population:adults aged 18 years or older with confirmed early-stage, nonmetastatic breast cancer (AJCC stages I–III) scheduled to receive, receiving, or having received systemic cytotoxic chemotherapy (neoadjuvant or adjuvant; anthracyclines, taxanes, alkylating agents and others). Combinations with endocrine, HER2-targeted or other therapies are eligible. Healthy controls and breast cancer patients without chemotherapy are eligible as comparators.

- Excluded: exclusively metastatic or stage IV disease, CNS metastases or primary brain tumours, pre-existing neurodegenerative disorders or major psychiatric illness, other active or recent (within 5 years) malignancies, mixed cancer or mixed-stage populations without separately extractable breast cancer stage I–III data, and non-cytotoxic modalities alone.

- Concept (The core phenomenon, intervention, or key variables of interest): relationship between circulating senescence/SASP biomarkers and cognitive outcomes in the context of cytotoxic chemotherapy.
    - Biomarkers (whole blood, serum, plasma or PBMCs): p16INK4a (CDKN2A), p21CIP1 (CDKN1A), p53, hypophosphorylated Rb, leukocyte telomere length, IL-6, TNF-α, IL-1β, IL-8, MCP-1/CCL2, GDF-15, PAI-1/SERPINE1; measured before, during, after or longitudinally following chemotherapy.
    - Cognitive outcomes: objective neuropsychological tests (e.g., CANTAB, CNS Vital Signs, HVLT-R, Trail Making Test, COWA, Stroop), validated patient-reported measures (e.g., FACT-Cog, PROMIS Cognitive Function) and CRCI incidence as defined by each study (e.g., ISCCC).
    - Comparisons when present: healthy controls, patients without chemotherapy, prechemotherapy baseline, different regimens. Studies without an external comparison group are eligible if they report within-subject change or correlation/regression analyses.
    - Also charted: fatigue, sleep, distress, depression and quality of life (e.g., FACT-F, BFI, PSQI, EORTC QLQ-C30, FACT-B).

- Context: Specific settings (geographic, cultural, healthcare settings, etc..
- Types of Sources: peer-reviewed primary human studies: prospective and retrospective cohorts, case-control and nested case-control, cross-sectional studies with quantitative association analyses, and RCTs or secondary analyses of trials. Grey literature, dissertations, preprints and conference proceedings. Excluded: preclinical, in vitro and animal studies; case reports, case series (N < 10), qualitative studies; narrative and systematic reviews, meta-analyses, consensus statements, protocols without results, editorials and letters; studies without standardised cognitive assessment; studies not quantifying senescence or SASP biomarkers.

Search Strategy:

- Information Sources: MEDLINE (via PubMed), Cochrane CENTRAL, Epistemonikos. Other sources: backward and forward citation tracking of included studies and relevant reviews.
- Search approach: JBI three-step search: (1) limited search in MEDLINE; (2) full search across all sources; (3) reference-list screening.
- Search Lines / Draft Query: four blocks joined byAND: population ("Breast Neoplasms" [MeSH], "breast cancer", "breast carcinoma"); exposure ("Antineoplastic Agents"[MeSH], "Chemotherapy, Adjuvant"[MeSH], "cytotoxic chemotherapy", "systemic chemotherapy"); biomarkers ("Cellular Senescence"[MeSH], "Cyclin-Dependent Kinase Inhibitor 2A" [MeSH], p16INK4a, CDKN2A, p21CIP1, IL-6, TNF-alpha, GDF-15, IL-8, IL-1β, "biological aging", "accelerated aging"); outcome ("Cognitive Dysfunction"[MeSH], "Chemotherapy-Related Cognitive Impairment"[MeSH], chemobrain, "chemo fog", CRCI, "executive function", memory, FACT-Cog). Terms within a block joined by OR. Full string with field tags for PubMed: ???
- Study Filters: no date restriction; English and Spanish; human studies; peer-reviewed. Deduplication before screening.

Source Selection (Screening Process):

- Records are de-duplicated and imported into a reference manager (Rayyen). Titles and abstracts are screened against the criteria, followed by full-text review of potentially eligible sources. Screening is performed independently by two reviewers; disagreements are resolved by consensus or a third reviewer. Reasons for excluding full texts are recorded.

Data Extraction (Data Charting):

- A list of variables to be extracted from each study (e.g., author, year, country, methodology, key findings).
- A draft or template of the data charting form.

Data Analysis and Synthesis:

- No meta-analysis, risk-of-bias assessment or certainty grading is planned, in line with JBI scoping review methodology. Results are presented as a PRISMA-ScR flow diagram, a numerical summary of included studies (year, country, design, regimen, sample size), and tables and charts mapping biomarker category and matrix, cognitive modality and domain, time window (baseline; during; up to 3 months; ≥6 months, with 3 to 6 months assigned to the nearest window and flagged), chemotherapy regimen and participant characteristics. Reported

associations are tabulated as published and summarised narratively to answer the review questions and identify gaps.

Study Management

Timeline:
Start date: 11th October 2026
End date: 15 th December 2026

Data Availability Statement:
The final extracted datasets will be made openly available on the OSF repository upon completion of the review.